Q1

$a = 4, b = 3/8, c = 2$

Q2

$x = [4 \pm \sqrt{(16 + 12(2+\sqrt{5})(2-\sqrt{5}))}] / [2(2+\sqrt{5})]$. This gives $x = (4 \pm \sqrt{(16-8)}) / (2(2+\sqrt{5})) = (4 \pm \sqrt{8}) / (2(2+\sqrt{5}))$

Q3a

leading coefficient = +3, one factor = $(x+2)$, remaining factors = $(x-2)(x+4)$

Q3b

Case 1: $5x-2 \geq 4x+1$ gives $x \geq 3$ (critical value 3). Case 2: $5x-2 \leq -4x-1$ oe. Combined solution: $x \leq 3$

Q4a

$r\theta = 12$ and $\frac{1}{2}r^2\theta = 57.6$. Solving: $r = 8, \theta = 1.5$

Q4b

$AC = 20$. Shaded area $= (\frac{1}{2} \times 20 \times 9.6) - 57.6 = 96 - 57.6 = 38.4$